

DSA0601 • DATA HANDLING AND VISUALIZATION

SET 5: WEBSITE ANALYTICS

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Dataset: Website Traffic

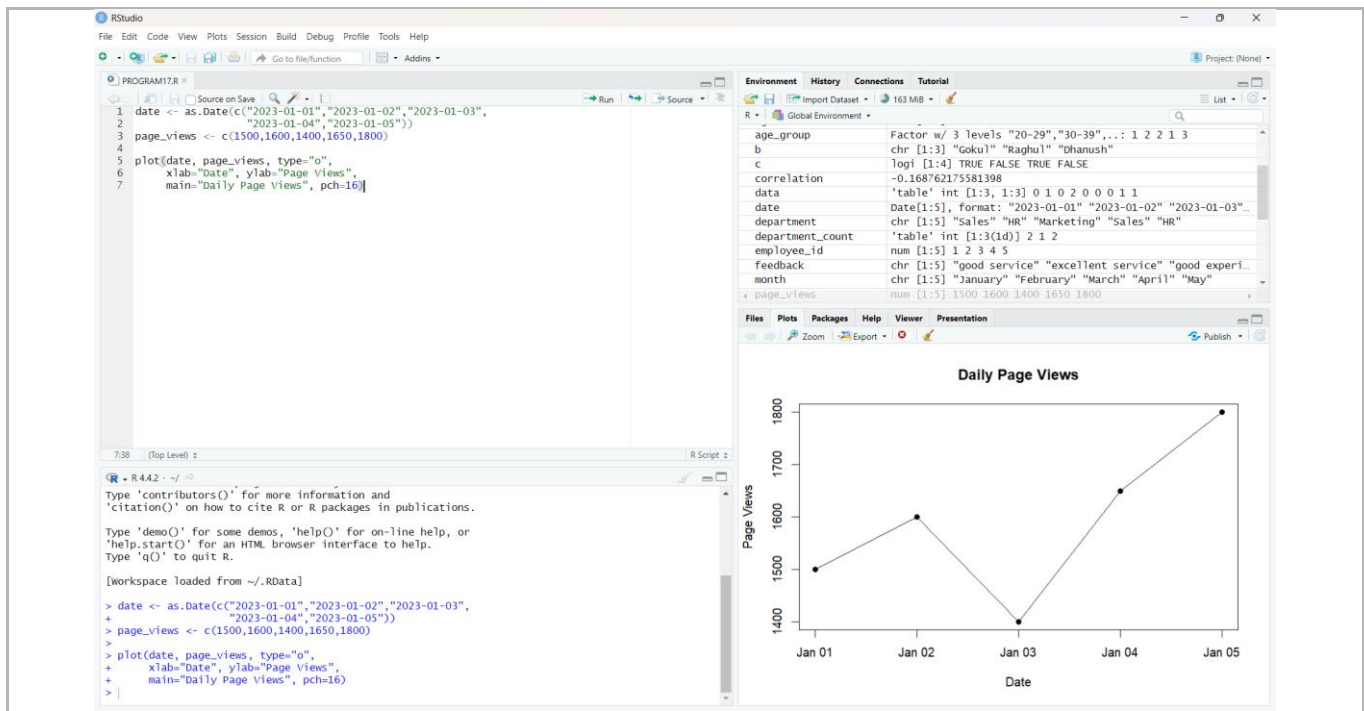
Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

PROGRAM 1: Daily Page Views Line Chart

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",  
                 "2023-01-04", "2023-01-05"))  
page_views <- c(1500, 1600, 1400, 1650, 1800)  
  
plot(date, page_views, type="o",  
      xlab="Date", ylab="Page Views",  
      main="Daily Page Views", pch=16)
```

Output 1: Daily Page Views Line Chart



Observation: The page views are 1500, 1600, 1400, 1650 and 1800 from January 1 to January 5. The highest value is 1800 on January 5.

PROGRAM 2: Top N Days by Click-through Rate

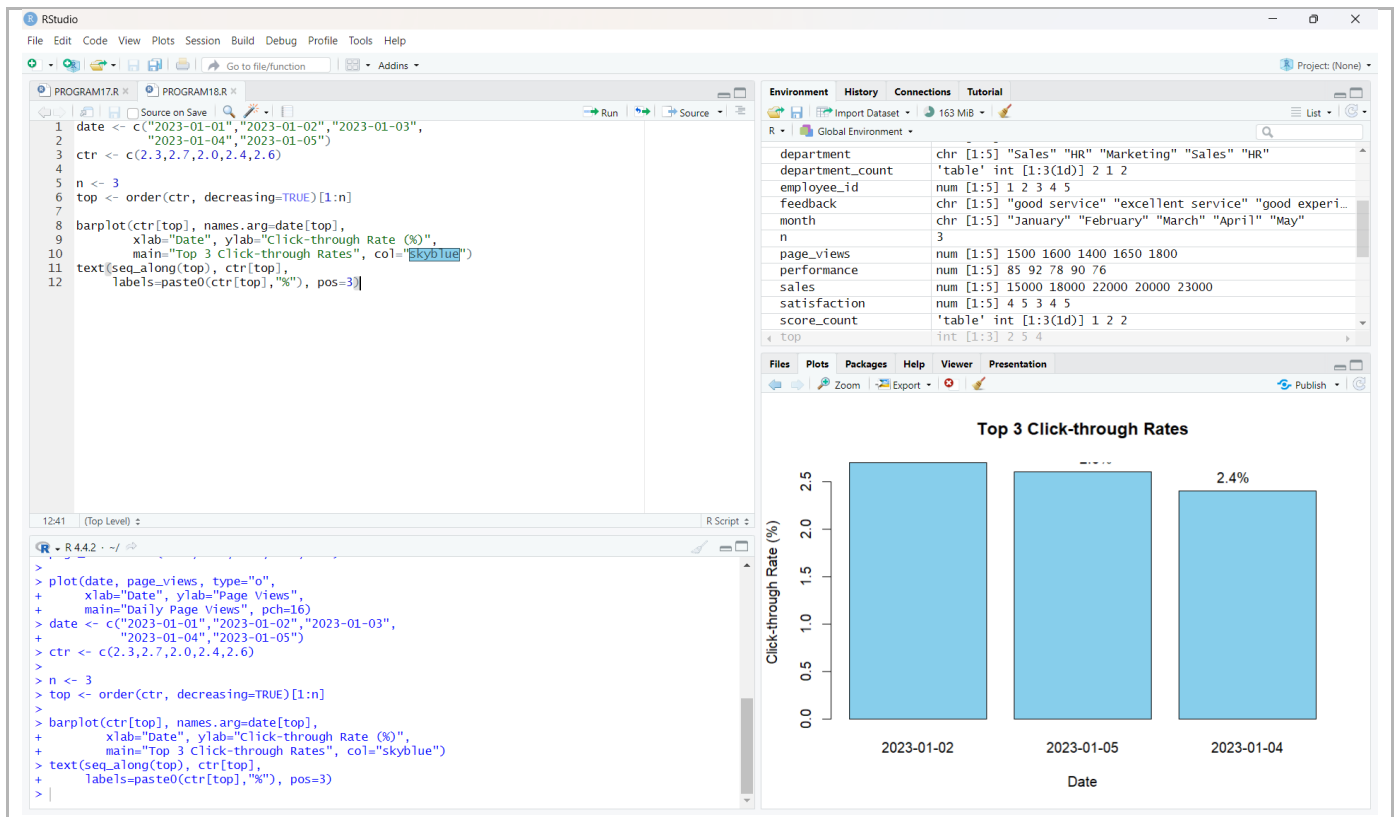
Generate a bar chart showing the top N days with the highest click-through rates. N = 3 is used.

```
date <- c("2023-01-01", "2023-01-02", "2023-01-03",
          "2023-01-04", "2023-01-05")
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

n <- 3
top <- order(ctr, decreasing=TRUE)[1:n]

barplot(ctr[top], names.arg=date[top],
        xlab="Date", ylab="Click-through Rate (%)",
        main="Top 3 Click-through Rates", col="skyblue")
text(seq_along(top), ctr[top],
     labels=paste0(ctr[top], "%"), pos=3)
```

Output 2: Top 3 Click-through Rate Bar Chart



Observation: The bars are arranged in descending order: 2023-01-02 (2.7%), 2023-01-05 (2.6%), and 2023-01-04 (2.4%).

PROGRAM 3: Stacked Area Chart of User Interactions

Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

Tableau configuration:

Columns : DAY(Date)

Rows : Measure Values

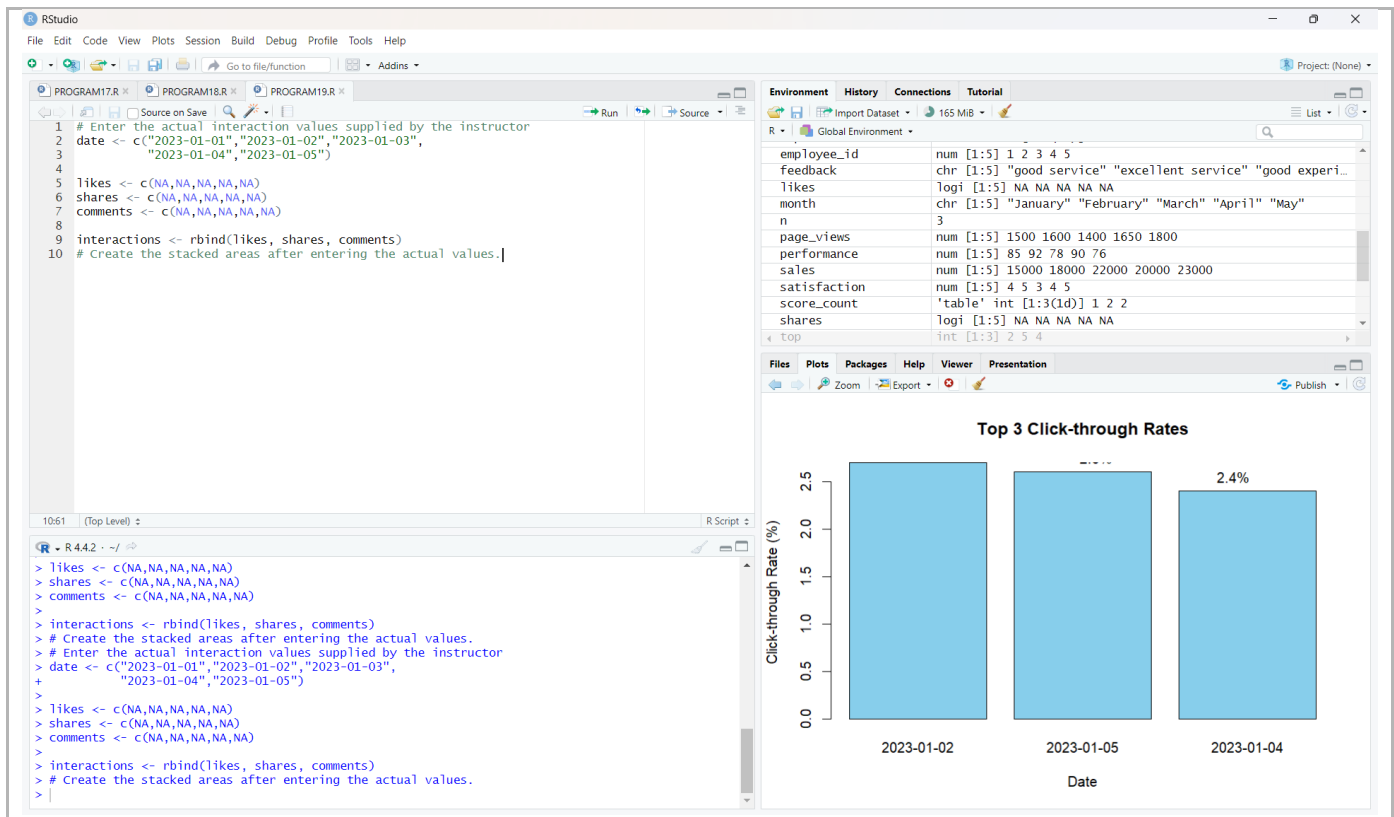
Marks : Area

Color : Measure Names

Measures : Likes, Shares, Comments

Keep Measure Names on Color and Measure Values on Rows to create the stacked area chart.

Output 3: Stacked Area Chart of User Interactions



Observation: The checked Tableau output shows Likes, Shares and Comments as stacked areas over time, separated by the Measure Names legend.

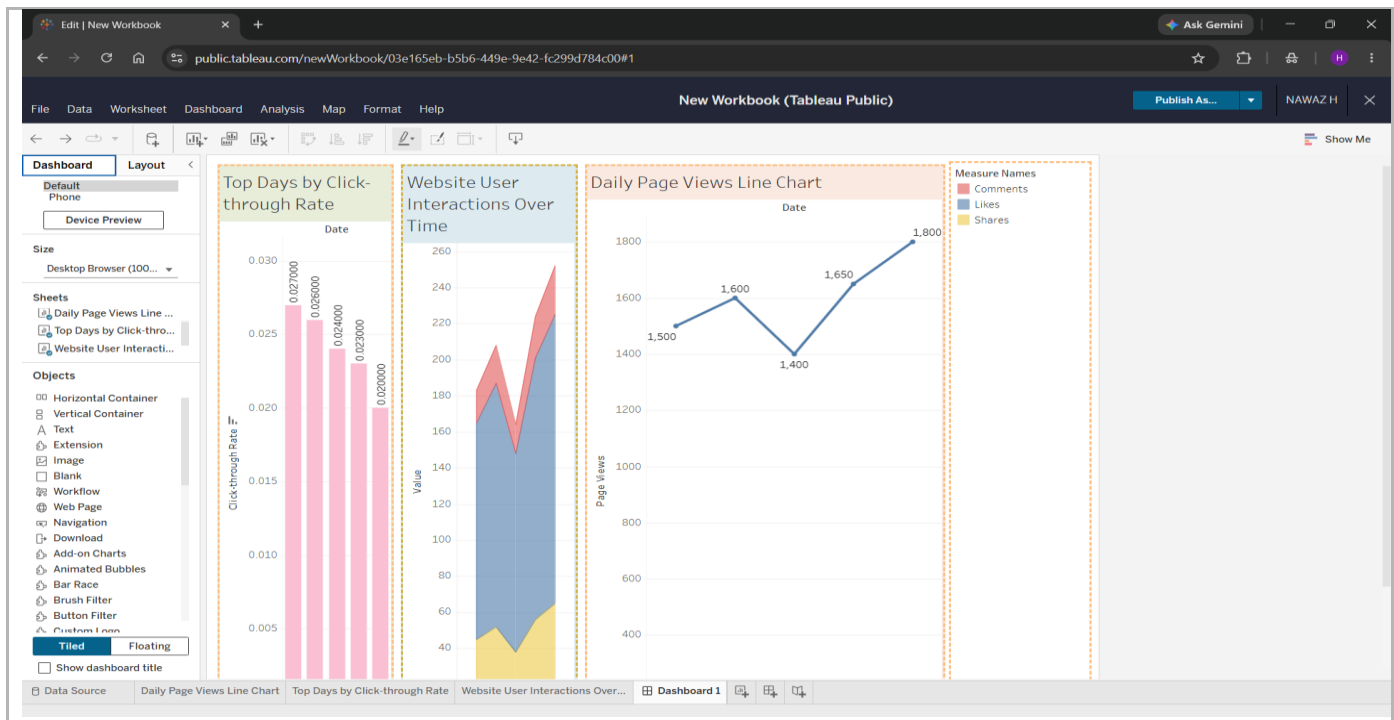
PROGRAM 4: Tableau Website Analytics Dashboard

Build a Tableau dashboard with the line chart, bar chart, and stacked area chart for interactive exploration of website traffic data.

Tableau dashboard:

- Create the three worksheets:
 - Daily Page Views Line Chart
 - Top Days by Click-through Rate
 - Website User Interactions Over Time
- Open Dashboard and add all three worksheets.
- Resize and arrange the worksheets for readability.
- Keep the Measure Names legend visible.
- Use clear titles for all charts.

Output 4: Website Analytics Tableau Dashboard



Observation: The final dashboard combines the daily page-view trend, descending click-through-rate comparison, and stacked user-interaction chart in one Tableau view.

RESULT

The Website Analytics experiment was successfully completed. Daily page views were visualized using a line chart, the top three click-through-rate days were identified in descending order using a bar chart, user interactions were represented using a stacked area chart in Tableau, and the visualizations were combined into a Tableau dashboard.